CityFix — Technical Documentation

SolveThe17 · Track 4: Unlocking Egypt’s Waste Economy

Submission snapshot: 2026-10-23. Software proof of concept with simulated hardware.

1. Purpose, users and honest scope

CityFix is the overall smart-city project, bringing together FixFirst (reporting), CityServe (municipal services), CityCycle (waste and recycling), CityOps (operations), CityPulse (maps) and CityBrain (AI assistance). **CityCycle is one aspect of CityFix** and the focus of this Track 4 waste-economy submission. It connects households offering sorted recyclables, informal collectors carrying out pickups, and district staff coordinating service. It adds waste recovery workflows to the CityFix citizen mobile app, citizen website and Command Center. The theme cites approximately 20 million tonnes of municipal solid waste annually in Egypt and recovery of up to 80% of the waste gathered by the Zabbaleen. The 80% figure is not Egypt's national recycling rate; these are challenge-provided facts, not our pilot results.

The implemented software includes pickup requests, collector acceptance, recorded material weights and cash amounts, household confirmation, bin telemetry ingestion, history and alerts. AI suggests; a person confirms. A deterministic fallback is not evidence of a successful Gemini call. Maps access does not establish backend access.

The PC simulator is operational locally; Wokwi bin and scale source and native logic tests are available. The physical solar/GSM bin is a paper design. No bench unit, physical scale accuracy, battery life or field deployment is claimed.

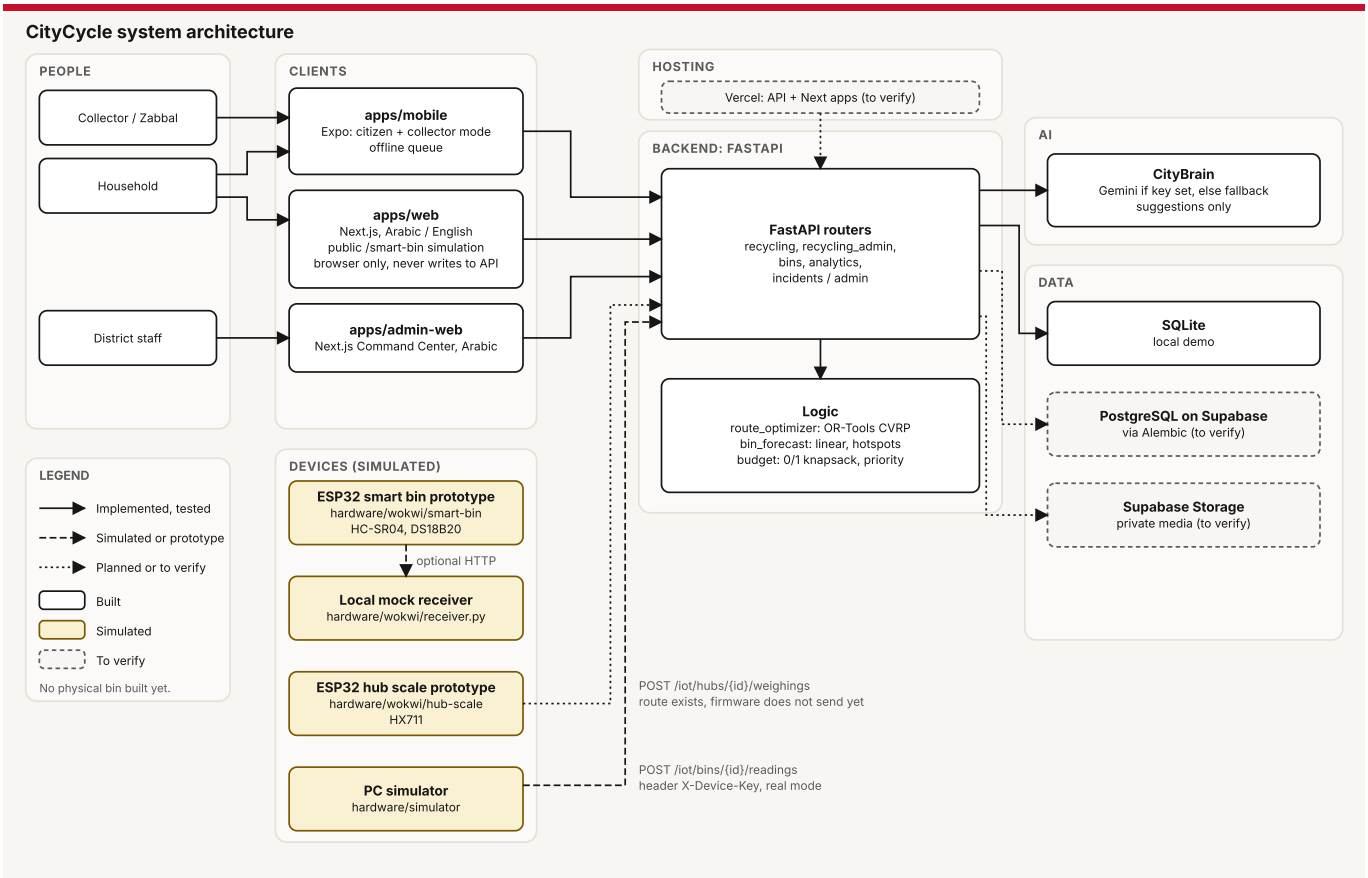
Proposed idea, not implemented acceptance: recycler handover receipts, confirmed recovery rather than collection alone, collector net earnings after costs, authenticated scale-to-pickup verification and weight/payment dispute handling. These extend the pitch and pilot plan; they are not features added by this documentation package.

1.1 CityFix modules and their status

Statuses follow docs/evidence.md. “Built” means the feature exists in the source and is covered by tests or screens; it does not mean field-tested or certified on the hosted deployment.

Module	What it does	Status today
FixFirst	Citizens report a problem with photo, voice note and location; an explainable priority score ranks it; staff plan repairs within a budget.	Built (mobile, web, Command Center). Keyword triage passes a 33-case synthetic smoke set; this is not an accuracy benchmark.
CityServe	Municipal service requests such as bulky-waste pickup, tree maintenance and garbage schedule.	Built in the citizen app; not field-tested.
CityCycle	Recyclables pickup requests, collector mode, district price list, weighings, household confirmation, impact ledger and smart-bin readings.	Built and verified in backend tests with demo prices and seed data. Smart bin and hub scale are simulated. Recycler receipts and confirmed recovery are proposed.
CityOps	Work orders and department assignment from the Command Center.	Built; tested with vitest, not run against a deployed API.
CityPulse	Live map of incidents, smart bins and dumping hotspots.	Built; bin readings come from simulators, not physical bins.
CityBrain	AI triage, fill forecasting, route optimisation and hotspot detection. AI only suggests; a person confirms.	Forecast, routing and hotspots verified in tests. Gemini and photo-analysis quality are to verify; photo analysis is off by default.

2. System architecture



The diagram describes implementation boundaries; hosting/database labels marked “to verify” are not deployment certifications. Team-provided web/API addresses and APK build are listed in the package index.

Component	Responsibility	Boundary / limitation
Expo mobile (`apps/mobile`)	Citizen and collector UI, maps, persistent queues	Mock by default in source; backend-linked preview build explicitly uses API mode. Native Maps configuration is separate from API configuration.
Citizen web (`apps/web`)	Household reporting, recyclables requests and tracking	Public browser smart-bin demo never writes to the API.
Command Center (`apps/admin-web`)	Staff pickup, bin, alert, payout and planning views	Requires backend URL and staff authorization.
FastAPI (`backend`)	Authentication, role checks, workflows, device ingestion, AI adapters	SQLAlchemy; local SQLite demo, PostgreSQL migration path. Live database state not certified.
CityBrain	Consented waste/photo suggestions, transcription adapters and fallback	Server-side Gemini key required for Gemini; live quality evaluation is still pending.
Hardware models	PC telemetry simulator, Wokwi ESP32 bin and hub scale	Scale delivery is local mock; no physical build.

Detailed architecture and API shapes: architecture, API contract, AI status.

3. Process flowcharts

Household / collector workflow

```
Household selects materials, location and pickup time
                          ↓
Backend stores request and price-list estimate (if prices exist)
                          ↓
Verified collector accepts → second acceptance is rejected
                          ↓
Collector records each material's weight and cash amount
                          ↓
Collector completes → household confirms → recorded impact
                          ↓
[PROPOSED] Recycler records accepted/rejected material and receipt
                          ↓
[PROPOSED] Confirmed recovery and collector net-earnings reporting
```

Collected kilograms are not proof of recycling. A recorded material value is not profit. A weighing labelled “scale” is not sufficient evidence of a calibrated, authenticated device measurement.

Smart-bin telemetry workflow

```
Sample fill distance / temperature / synthetic battery
                          ↓
Validate sensors → invalid reading: show fault, do not invent data
                          ↓
Queue immutable reading with increasing sequence number
                          ↓
Offline / failed send? → retain reading and retry with backoff
                          ↓
Receiver authenticates device key and de-duplicates (bin ID, seq)
                          ↓
202 + accepted sequence IDs → remove acknowledged rows only
                          ↓
Backend latest state/history → Command Center and citizen bin layer
```

The PC simulator can target the real API. Wokwi defaults to local mock; optional bin HTTP needs explicit endpoint/key/TLS configuration. Wokwi scale HTTP is not implemented. These are separate demonstrations, not an automatic connected chain.

4. Device contract and security

`POST /iot/bins/{id}/readings` uses `X-Device-Key`, a batch of readings and a `202` acknowledgement. Example values are illustrative:

```
{"readings":[{"seq":1042,"at":null,"fillPct":82,"distanceCm":22,
"tempC":38.5,"batteryV":3.91,"rssi":-85,"fw":"sim-0.1.0"}]}
```

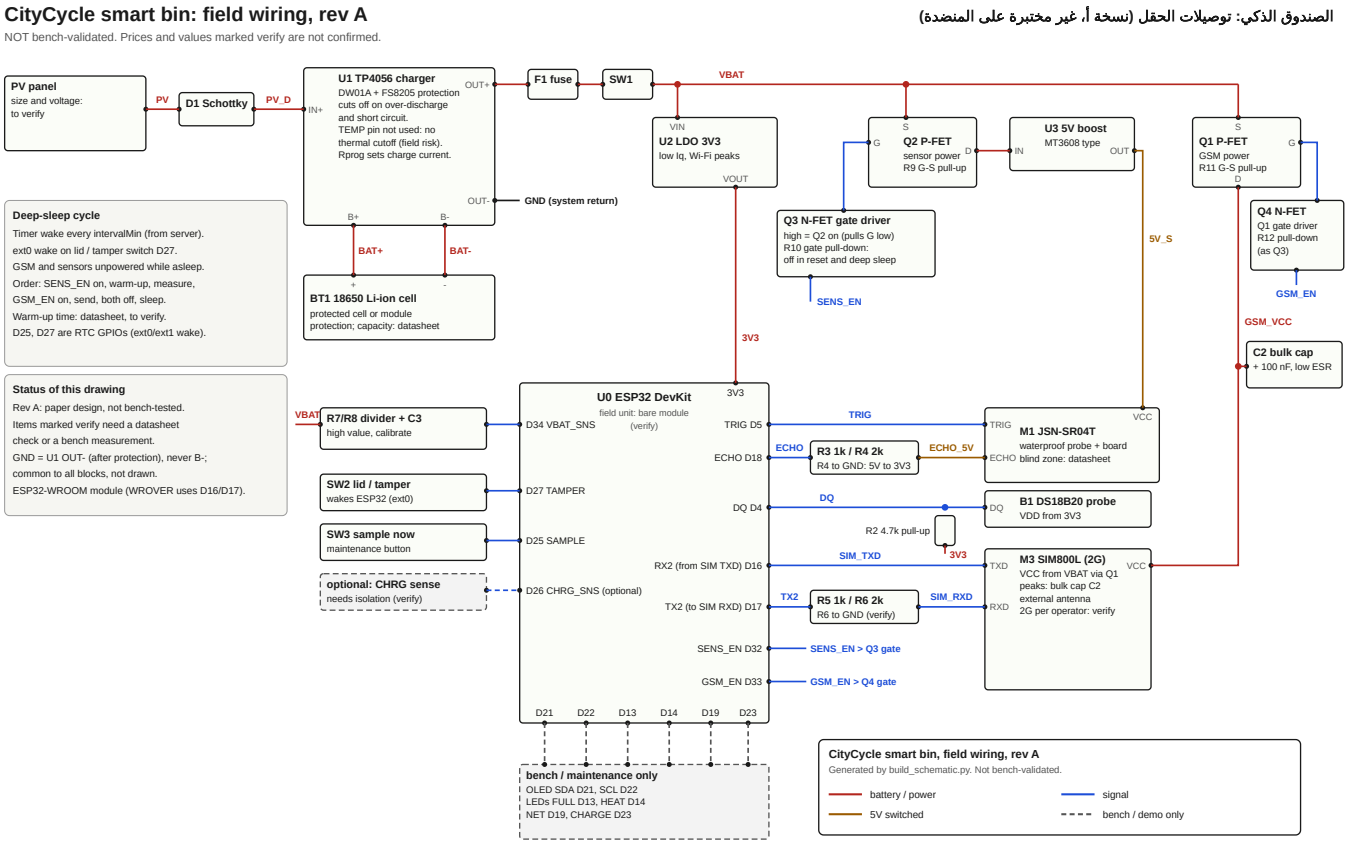
```
{"accepted":[1042],"intervalMin":30,"fullPct":80,"heatC":60}
```

An unknown clock is stamped by the receiver and marked unsynced. Duplicate identical sequences are safe; conflicting reuse is rejected. Policies change only after a valid acknowledgement. Device registry/rotation, staff roles and citizen tokens are separate identities; keys must never appear in screenshots or committed files.

`POST /iot/hubs/{id}/weighings` exists in the backend and acknowledges a sequence. The Wokwi scale has a persistent local queue but does not send HTTP. Connecting this measurement reliably to a collector pickup remains integration work. Cash is recorded, not electronically transferred.

Citizen photos need consent for model analysis and private upload configuration. Gemini credentials belong on the server. Exact household locations and collector financial records need role-restricted access. Retry/idempotency and offline queues reduce duplicate writes; they do not establish delivery without an acknowledgement.

5. Circuit and field design



The separately supplied **CityFix-Circuit-Diagram.pdf** provides a larger A3 view. The SVG and generated pin table are in schematics.

Block	Proposed field component	Engineering requirement still to validate
Controller	ESP32	Real deep-sleep firmware and measured whole-board sleep current
Fill	Waterproof JSN-SR04T or suitable alternative	Mounting blind zone, per-bin calibration, voltage levels and contamination
Temperature	Waterproof DS18B20	Placement and response time; bin temperature is not battery thermal protection
Street connectivity	SIM800L, subject to 2G availability; LTE alternative	Operator service, modem current peaks, external antenna and TLS capability
Power	Solar, protected 18650 cell, charger, fuse and regulator	Solar sizing, cell charge-temperature cutoff, safe power path and discharge test
Enclosure	Gasketed enclosure, glands, mounting and security screws	Assembled dust, splash, heat and maintenance tests; no verified IP rating claimed
Optional hub scale	Load cell + HX711 + ESP32	Calibration, stability, payload range and trusted backend/phone handover

The Wokwi battery potentiometer and solar switch are synthetic controls, not an energy model. A real battery needs a divider; a 5 V sensor echo must not go directly to the ESP32. The design notes identify a charger without thermal cutoff and unverified solar-input compatibility: component substitution and bench validation are gates before a physical deployment.

6. Estimated Bill of Materials in EGP

Use CityFix-BOM-Estimate.csv with COST-ESTIMATE.md. The budget is based on dated Egyptian supplier prices, with item quantities and source links. Prices verify a budget input, not suitability of the circuit. One custom mounting row still needs a workshop quote; it is not silently counted as free.

Cost scope	EGP
Smart-bin priced components subtotal (custom mount excluded)	7,243.00
Calendar-month data reference, derived from 37 EGP/28 days	40.19
Custom lid / sensor mount	Workshop quote required

Totals are generated into the cost report. They separate the priced smart-bin component subtotal, monthly connectivity reference and optional scale component subset. Existing computer/phone/bin assets, labour, collection vehicles, installation, cloud/provider usage and taxes/delivery not included by a supplier require separate budgets. Replacing 2G or the charger/power design can materially change cost.

7. Prototype evidence and reproduction

The prototype runbook provides a dependency-free PC demonstration and a full local software setup. The verification record records the commands actually executed for this package and distinguishes tests from live service verification.

The local simulator tests cover queue persistence, retry handling, receiver behaviour and controls. Native C++ tests cover bin and scale logic. Backend integration tests exercise authenticated bin registration/ingestion/history and collector workflows using a test database, not the hosted database.

The team supplied hosted addresses for all three web services and reported Maps working on an Android APK. Workspace network policy prevented independent live verification. Gemini requires a valid server-side key and supported media path; accuracy, production provider configuration and live OTP must be checked separately. No database migration or production data write was performed for this package.

8. Egyptian deployment and evaluation

Submit the separate two-page **CityFix-Field-Deployment-Brief.pdf**. It covers heat, dust, street washing, weak networks, power, vandalism, low literacy, worker participation, maintenance and pilot gates.

Start with one neighbourhood, participating collectors, a hub and a recycler. Retrofit existing assets where useful. Measure collection time, overflow duration, contamination and cost per collected kilogram. Confirmed recycler recovery and collector net earnings require the proposed receipt/cost tracking, not current collection totals. Set numerical success targets with partners after a baseline; no unmeasured benefit, accuracy, payback or emissions saving is claimed.

References: challenge framing, claims and evidence, field engineering, dated supplier evidence, power formulas, presentation.

Appendix: estimated EGP budget and line items

BOM with sourced EGP prices

Prepared 2026-10-23. The table uses published Egyptian supplier prices for the exact item or set specified in the CSV. It replaces the earlier planning allowances. Prices are evidence for a budget, not proof that the circuit is field-ready. See dated price research.

Cost summary

Cost scope	EGP
Smart-bin priced components subtotal (custom mount excluded)	7,243.00
Calendar-month data reference, derived from 37 EGP/28 days	40.19
Custom lid / sensor mount	Workshop quote required

Scope

The BOM has 26 priced rows and one custom-work row still requiring a quote. The optional LTE alternative is excluded from the bin subtotal. No full installed-cost or payback claim is made. Component specifications and the solar/power circuit still need validation. The source register records which figures were read from product pages, indexed supplier listings or derived from multiple published prices.

The 37 EGP Vodafone reference renews every 28 days: 37 × 365 ÷ 28 ÷ 12 = 40.19 EGP per average calendar month. Activation, business IoT terms and applicable recharge charges need operator confirmation. This is not a committed IoT service offer.

Delivery, taxes not included by a supplier, installation, labour, vehicles, existing bin/phone/computer assets, calibration, further wiring and power-design substitutions are outside the component subtotal. The custom mount has no verifiable fixed price until dimensions and fabrication method are agreed; it is not counted as free.

Itemized component prices

ID	Item	Quantity	Unit EGP	Total EGP	Supplier
B01	ESP32 dev board	1	280.00	280.00	Makers Electronics
B02	JSN-SR04T ultrasonic probe	1	375.00	375.00	Electra Store
B03	DS18B20 waterproof probe	1	65.00	65.00	Makers Electronics
B04	Pull-up resistor 4.7k	1	0.25	0.25	Makers Electronics
B05	Resistor assortment kit	1	95.00	95.00	Makers Electronics
B06	TP4056 charger module	1	15.00	15.00	Makers Electronics
B07	18650 Li-ion cell	1	165.00	165.00	Makers Electronics
B08	18650 single-cell holder	1	8.00	8.00	Makers Electronics
B09	Solar panel	1	240.00	240.00	Makers Electronics
B10	Schottky reverse-protection diode	1	1.25	1.25	Makers Electronics

ID	Item	Quantity	Unit EGP	Total EGP	Supplier
B11	Inline fuse and holder	1	10.75	10.75	RAM Electronics
B12	3.3 V LDO low quiescent	1	16.00	16.00	RS Egypt
B13	5 V boost module	1	25.00	25.00	Makers Electronics
B14	P-channel MOSFET	2	3.50	7.00	Makers Electronics
B15	Gate-drive transistor	2	2.50	5.00	UGE Electronics Egypt
B16	Bulk capacitor	1	2.00	2.00	Makers Electronics
B17	SIM800L GSM module with antenna	1	300.00	300.00	Makers Electronics
B18	LTE Cat-1 modem module	1	10000.00	10000.00	Makers Electronics
B19	SIM card and data plan	1	40.19	40.19	Vodafone Egypt
B20	Weatherproof enclosure	1	275.00	275.00	Makers Electronics
B21	Cable gland PG7 nylon	3	4.00	12.00	Makers Electronics
B22	Breather vent plug	1	389.85	389.85	AETRONIX Egypt
B23	Security screws	1	1883.00	1883.00	Desertcart Egypt
B24	Lid bracket and sensor mount	1	Quote required	—	Custom workshop
B25	Conformal coating	1	2352.90	2352.90	AETRONIX Egypt
B26	Silica gel sachets	1	135.00	135.00	Silica Egypt
B27	Wiring and connectors	1	540.00	540.00	Makers Electronics and RAM Electronics
B28	Perfboard or small PCB	1	45.00	45.00	RAM Electronics

Optional hub-scale reference

The current Wokwi scale is a simulation. Published component references: ESP32 280 EGP, 50 kg bar load cell 350 EGP, HX711 75 EGP, giving a **705 EGP component subset**, not a complete scale. Platform fabrication, supply, display/controls, calibration and integration still need a design and quote. See the research register for supplier URLs. The HX711 product-page reference is used rather than a conflicting category price.

Completing the only unpriced bin row

B24 is a custom lid bracket and sensor mount. A real value requires the selected bin dimensions, a finished mounting drawing and a local workshop quotation. No unrelated sensor bracket is substituted and no fabricated price is entered. Obtain that quote before claiming a complete installed-hardware budget.